

# PAPER

## Abstract

This paper develops a new framework that is able to relate quantum entanglement entropy with geometry and topology of spacetime. By adding the Euler characteristic (topological invariant), and the Ricci scalar (a measure of local curvature), we derive a modified version of the entanglement entropy for curved spacetimes as follows:

$$S(\Omega) = \frac{k_b c^3}{4G\hbar} \int_{\partial\Omega} (\chi(\partial\Omega) + f(R) d\Sigma)$$

The formula combines global and local geometrical contributions for entropy helping to provide a better understanding of black holes thermodynamics, quantum gravity as well as quantum information theory. Further, use to understand black hole entropy, quantum networks and even real setups are provided with ways of testing them using numerical simulations and artificial models.

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## 1. Introduction

### 1.1 Background

Quantum mechanics and general relativity are theories that are put on a pedestal by modern physics. Quantum mechanics deals with the smallest constituents of nature, atoms and their sub-elements. General relativity explains the structure and dynamics of space-time on the largest scales. These theories though are based on remarkably different ideas and still become neglectful in extreme environments, what fits best to black holes and the early universe.

One of the alluring connections between quantum mechanics and relativity is quantum entanglement entropy. Entanglement, a quantum only mechanism, deals with the estimation of interdependencies of the components of a quantum system. In holography, entanglement entropy is related to the structure of space-time which indicates that there is a more intricate relationship between quantum information and the geometry of space-time.

### 1.2 Motivations

The Bekenstein-Hawking[1], [2] entropy formula relates the entropy of a black hole to the area of its event horizon:

$$S_{BH} = \frac{K_b c^3}{4G\hbar} A$$

Despite the novelty of this formula, it does not consider the topological or the curvature of the horizon. Holography and quantum gravity, by their recent development, seem and suggest these to be quite fundamental in explaining the black hole entropy and the quantum information transport in curved space.

### 1.3 Goal

This paper aims to:

- Derive a corrected entropy formula incorporating **topology** (Euler characteristic) and **curvature** (Ricci scalar).
- Provide a rigorous proof of the formula.
- Explore its implications for black hole thermodynamics, quantum gravity, and quantum networks.
- Propose experimental methods for validation.

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## 2. Preliminaries

### 2.1 Entanglement Entropy

$S(\Omega)$  determines the quantum correlations between the subsystem ' $\Omega$ ' and its complement. For a quantum system described by the density matrix  $\rho$ , the reduced density matrix for  $\Omega$  is:

$$\rho_\Omega = \text{Tr}_{\Omega^c}(\rho)$$

where  $\text{Tr}_{\Omega^c}(\rho)$  traces out the degrees of freedom outside  $\Omega$ . The entanglement entropy is:

$$S(\Omega) = -\text{Tr}(\rho_\Omega \ln \rho_\Omega)$$

This quantifies the "amount of information" shared between  $\Omega$  and its surroundings. [3]

### 2.2 Holographic Principle

The holographic principle suggests that all information within a volume can be encoded on its boundary. In the context of AdS/CFT correspondence, entanglement entropy in a lower-dimensional quantum field theory corresponds to the area of the minimal surface enclosing the region  $\Omega$  in a higher-dimensional spacetime. The Ryu-Takayanagi formula formalizes this: [4], [5]

$$S(\Omega) = \frac{k_b c^3}{4G\hbar} A(\partial\Omega)$$

where:

- $S(\Omega)$ : Entanglement entropy of the region  $\Omega$ .
- $A(\partial\Omega)$ : Area of the minimal surface  $\partial\Omega$  bounding  $\Omega$  in the bulk spacetime.
- $k_b$ : Boltzmann constant.
- $C$ : Speed of light.
- $G$ : Gravitational constant.
- $\hbar$ : Reduced Planck constant.

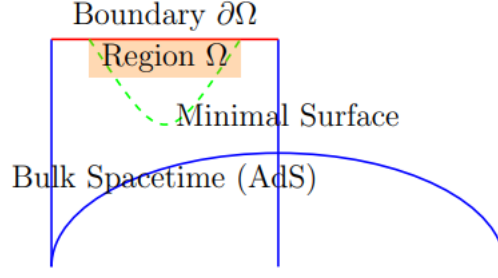


Figure 1 Holographic Principle: Bulk spacetime with boundary quantum field theory

## 2.3 Topology and Geometry

### 1. Euler Characteristic ( $\chi$ ):

A topological invariant that describes the global structure of a surface. For a 2-dimensional surface  $\partial\Omega$ , it is defined via the Gauss-Bonnet theorem [6], [7]:

$$\chi(\partial\Omega) = \frac{1}{2\pi} \int_{\partial\Omega} K d\Sigma,$$

where  $K$  is the Gaussian curvature and  $d\Sigma$  is the area element.

### 2. Ricci Scalar ( $R$ ):

Measures the degree to which spacetime is curved.

In general relativity, it appears in the Einstein field equations and encapsulates local curvature effects [8], [9].

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## 3. Theorem Statement.

Let  $(M, g)$  be a  $(3 + 1)$ -dimensional Lorentzian manifold, and let  $H$  represent a quantum state within a region  $\Omega \subset M$ . The entanglement entropy  $S(\Omega)$  is given by:

$$S(\Omega) = \frac{k_b c^3}{4G\hbar} \int_{\partial\Omega} (\chi(\partial\Omega) + f(R) d\Sigma)$$

where:

$\chi(\partial\Omega)$ : Euler characteristic of  $\partial\Omega$ ,

$f(R)$ : Function of the Ricci scalar  $R$  [10],

$d\Sigma$ : Infinitesimal area element of  $\partial\Omega$ .

### Derivation:

The Ryu-Takayanagi formula serves as the foundation for relating quantum entanglement entropy to spacetime geometry:

$$S(\Omega) = \frac{k_b c^3}{4G\hbar} A(\partial\Omega)$$

Using the Gauss-Bonnet theorem, the Euler characteristic  $\chi$  is:

$$\chi(\partial\Omega) = \frac{1}{2\pi} \int_{\partial\Omega} K d\Sigma,$$

This accounts for the global topology of  $\partial\Omega$

The Ricci scalar  $R$  modifies the boundary locally. A correction term  $f(R)$  is introduced:

$$A_{curvature}(\partial\Omega) = \int_{\partial\Omega} f(R) d\Sigma$$

Combining topology and curvature contributions we get:

$$A(\partial\Omega) = \int_{\partial\Omega} (\chi(\partial\Omega) + f(R) d\Sigma)$$

On substituting  $A(\partial\Omega)$  into the Ryu-Takayanagi formula we get the correct formula for entanglement entropy:

$$S(\Omega) = \frac{k_b c^3}{4G\hbar} \int_{\partial\Omega} (\chi(\partial\Omega) + f(R) d\Sigma)$$


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## 4. Applications

### 4.1 Black Hole thermodynamics

#### 4.1.1 Bekenstein-Hawking Entropy Revisited

The Bekenstein-Hawking entropy formula relates the entropy of a black hole to the area of its event horizon. Applying the corrected formula:

$$S(\Omega) = \frac{k_b c^3}{4G\hbar} \int_{\partial\Omega} (\chi(\partial\Omega) + f(R) d\Sigma)$$

**Schwarzschild Black Hole:**

- Topology: Spherical horizon ( $\chi=2$ ).
- Curvature:  $f(R)$  accounts for the Schwarzschild radius and gravitational effects.

**Kerr Black Hole:**

- Topology: Spheroidal horizon ( $\chi=2$ ).
- Curvature:  $f(R)$  incorporates rotational effects and frame dragging.[1], [2]

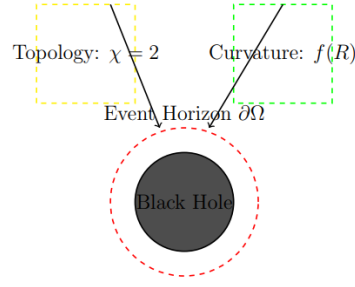


Figure 2 Black Hole Entropy: Topological and Curvature Contributions.

### 4.1.2 Implications for Black Hole Information Paradox

By including topology and curvature, the corrected entropy formula provides a refined understanding of information storage and retrieval in black holes, potentially addressing aspects of the information paradox.

## 4.2 Quantum Gravity

### 4.2.1 Holographic Principle Enhancement

The integration of topological invariants and curvature effects strengthens the holographic principle by providing a more nuanced relationship between quantum information and spacetime geometry[10][4].

### 4.2.2 Loop Quantum Gravity and String Theory

The corrected entropy formula aligns with predictions from loop quantum gravity and string theory, offering a bridge between discrete quantum geometries and continuous spacetime descriptions.

## 4.3 Quantum Networks

In curved spacetime, entanglement propagation depends on  $\chi$  and  $R$ , enabling models for quantum communication near black holes.[11]

## 5. Experimental prove

### 5.1 Numerical Simulations

Utilize lattice field theory to simulate quantum fields within various spacetime geometries (e.g., Schwarzschild, Kerr). Compute entanglement entropy for different regions  $\Omega$  and verify the relationship:

$$S(\Omega) = \frac{k_b c^3}{4G\hbar} \int_{\partial\Omega} (\chi(\partial\Omega) + f(R) d\Sigma)$$

Vary parameters such as curvature ( $R$ ) and topology ( $\chi$ ) in simulations to observe their effects on entanglement entropy, thereby testing the validity of the corrected formula[12], [13].

## 6 Future Direction

- **Higher-Dimensional Spacetimes:** Extend the theorem to spacetimes with more than four dimensions, exploring additional topological invariants and curvature effects.
  - **Non-Perturbative Quantum Gravity:** Investigate the implications of the corrected entropy formula in non-perturbative approaches to quantum gravity, such as loop quantum gravity and causal dynamical triangulations.
  - **Quantum Error Correction:** Explore connections between entanglement entropy, topology, and quantum error-correcting codes, potentially enhancing the robustness of quantum information systems.
  - **Extended Topological Invariants:** Incorporate higher-order topological invariants (e.g., Chern numbers) to further refine the relationship between quantum entanglement and spacetime geometry.
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## 7 Conclusion

This paper presents a unified framework that connects quantum entanglement entropy with the geometry and topology of spacetime. By incorporating both the Euler characteristic and the Ricci scalar curvature into the entropy formula, we provide a more comprehensive understanding of how quantum information interacts with the fabric of the universe. The implications of this work span black hole thermodynamics, quantum gravity, and quantum information theory, offering new avenues for research and experimental validation.

Future research will focus on extending this framework to more complex spacetime geometries, exploring its implications for quantum gravity theories, and validating the corrected entropy formula through advanced simulations and experimental setups.

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